NBA Shots - 4.4 Million Stories
Process book - COM-480 Project - Milestone 3

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Repository: https://github.com/com-480-data-visualization/com-480-project-pff
Live site:

Introduction

The NBA has been quietly rewritten over the last twenty years. The mid-range jumper which once was the heart of the game is now nearly extinct and was replaced by a flood of three-pointers and shots at the rim. Our project, NBA Shots - 4.4 Million Stories, is a single-page site built on every regular-season field-goal attempt from the 2003-04 season through 2024-25. It contains 4,444,820 shots across 22 seasons, 30 franchises, and 2,265 players. The site guides viewers through a single narrative arc: how does an NBA player decide to take a shot? Through seven dynamic and engaging visualizations, we explore how era, team context, and pressure shape the shot selection of the world's greatest basketball players and athletes.
This document describes how we got from the original concept (Milestone 1) to that final product. It covers the design decisions we made along the way, the challenges we ran into, and which member of the team did what.

From sketch to site: how the concept evolved

The original Milestone 1 / Milestone 2 plan

In Milestone 2 we proposed a minimum viable product with four core pieces:

1. Spatial shot evolution over time (how shot locations change across seasons)
2. Player shot profiles (graphical signatures with summary statistics)
3. Court value map (where on the floor the best shots are)
4. Clutch-time shots (the last five seconds of Q4 with team / player filters)

We also flagged two stretch goals: (5) team-level shot identities and (6) how a player's shot profile changes after a trade.

What changed between M2 and M3

The final site keeps all four MVP pieces and both stretch goals, but the order and framing changed substantially.

The court value map was promoted to the opening premise of the site. Every other visualization only makes sense once we understand that some parts of the court are mathematically more valuable than others, so it now anchors the narrative as Section 1.

The spatial shot evolution grew into the centerpiece as Section 2. We added a hexbin map for each year's league average, an animated season slider, and a companion trend chart tracking three-point, mid-range, and at-rim rates.

The team-level shot identities which was originally a stretch goal became Section 3. The key addition was a champion-comparison overlay, letting view

ers understand which strategies actually won.

The player shot profiles were entirely reframed in Section 4. Instead of conventional bar charts, shots became radial fingerprints. The shape is the player's signature, instantly comparable across players in a way flat charts never could be.

The post-trade shot evolution which also was originally a stretch goal became Section 5. We focused on four high-profile movers (LeBron, Durant, Harden, and Westbrook) to ask whether shot identity belongs to the player or to the system around them.

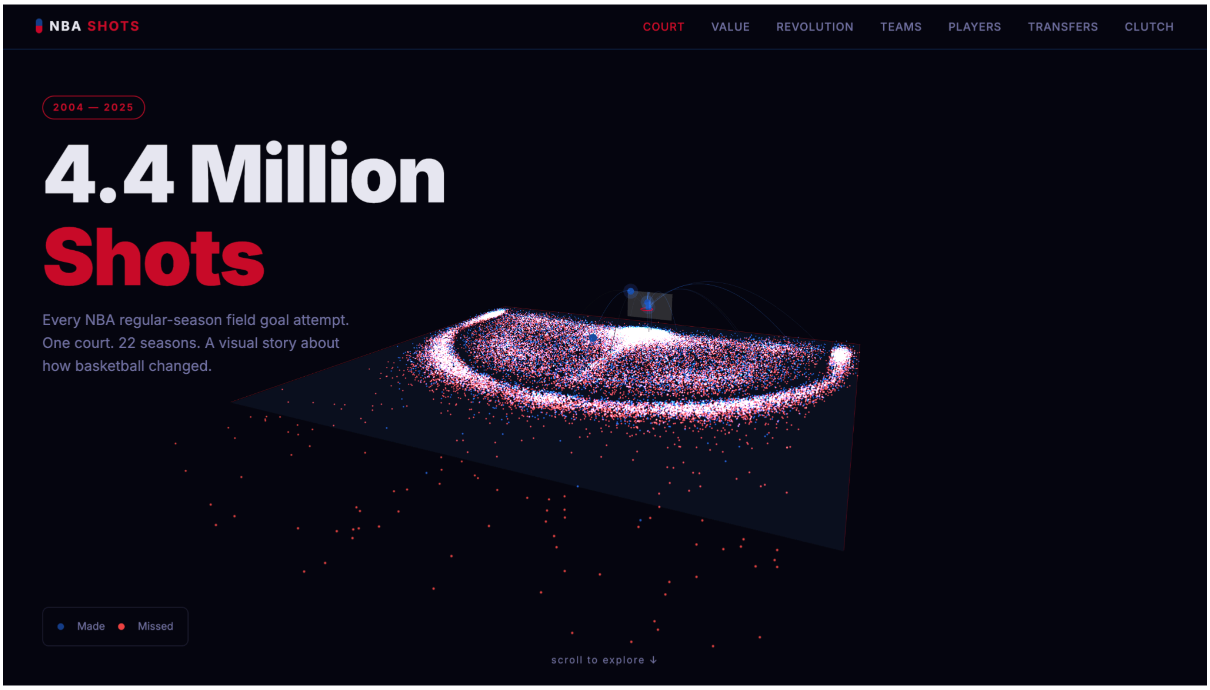
Finally, Section 6 closes the report by examining clutch-time shot selection. However we changed it to be not in isolation, but against the baseline of how shots are typically taken. We switched to a heatmap comparing clutch frequency to normal shot profiles after finding it gave a much clearer picture of how high-stakes moments reshape decision-making on the court.

The biggest change between milestones was structural, not technical. The goal was to ensure more story telling and not solely focus on the visualisations.

The final visualisation, section by section

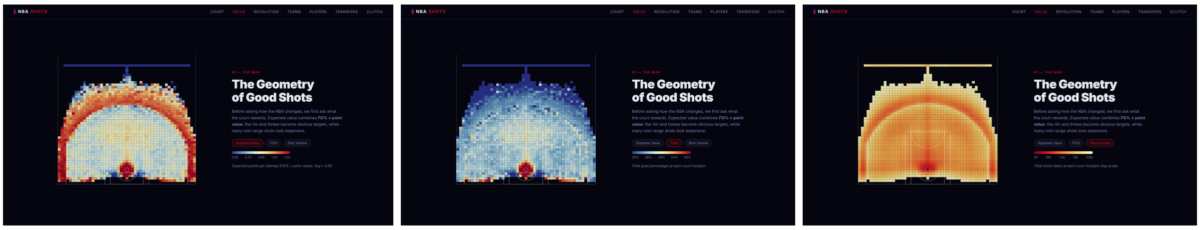
Hero - 4.4 Million Shots

A WebGL particle system raining one dot per shot onto the half-court. The made shots are shown in blue, missed in red, staggered fall using a custom GLSL shader for the ease-out animation. We used Three.js and not D3 as rendering ~75,000 sampled particles smoothly in SVG is not viable. WebGL with GPU-side animation runs comfortably at 60 fps.



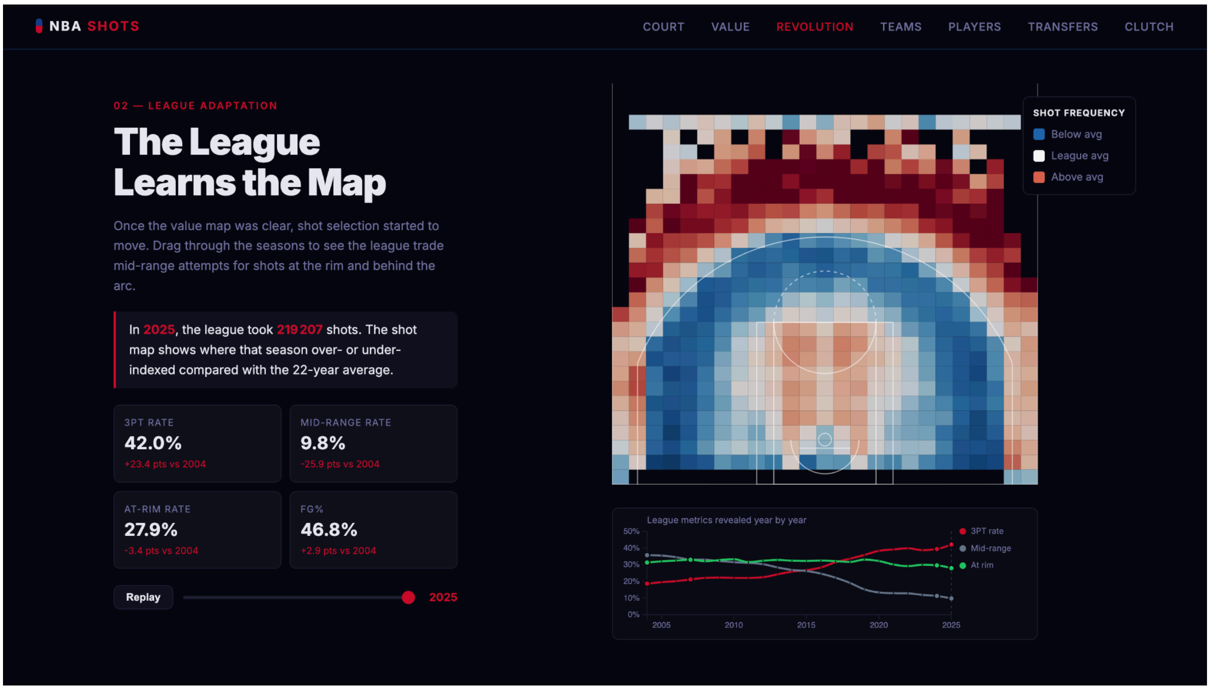
Value - The Geometry of Good Shots

A grid-cell heatmap with three toggleable encodings: Expected Value (FG% × points), FG%, and Shot Volume. This is the base to the whole website as it shows that the court is not uniform. The rim and the three-point line glow because they pay more per attempt.
We used a divergent red-grey scale centered at the league average (~0.93 EV per attempt) rather than a sequential scale, so users see deviation from the norm rather than absolute values.



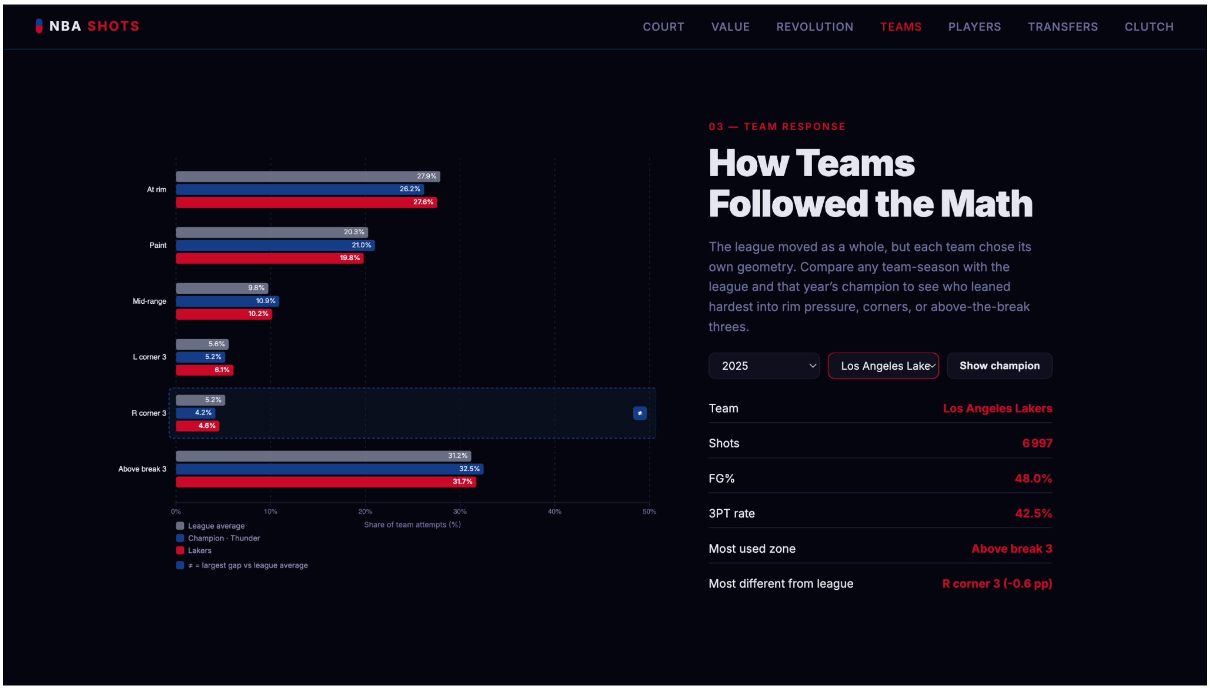
Revolution - The League Learns the Map

A hexbin map shows where shots came from in a given season, coloured by frequency relative to league average that year. The slider and the play button allow the user to watch the map mutate from 2004 to 2025. A small trend chart underneath tracks 3PT-rate, mid-range rate, and at-rim rate over time.
We decided to use hexbins, not raw scatter as at 200k+ shots per season, scatter is unreadable when hexbins aggregate cleanly. The divergent palette anchored on the league average emphasises shift and not absolute volume. Having an animation instead of a toggle allows the viewer to better grasp the insight.



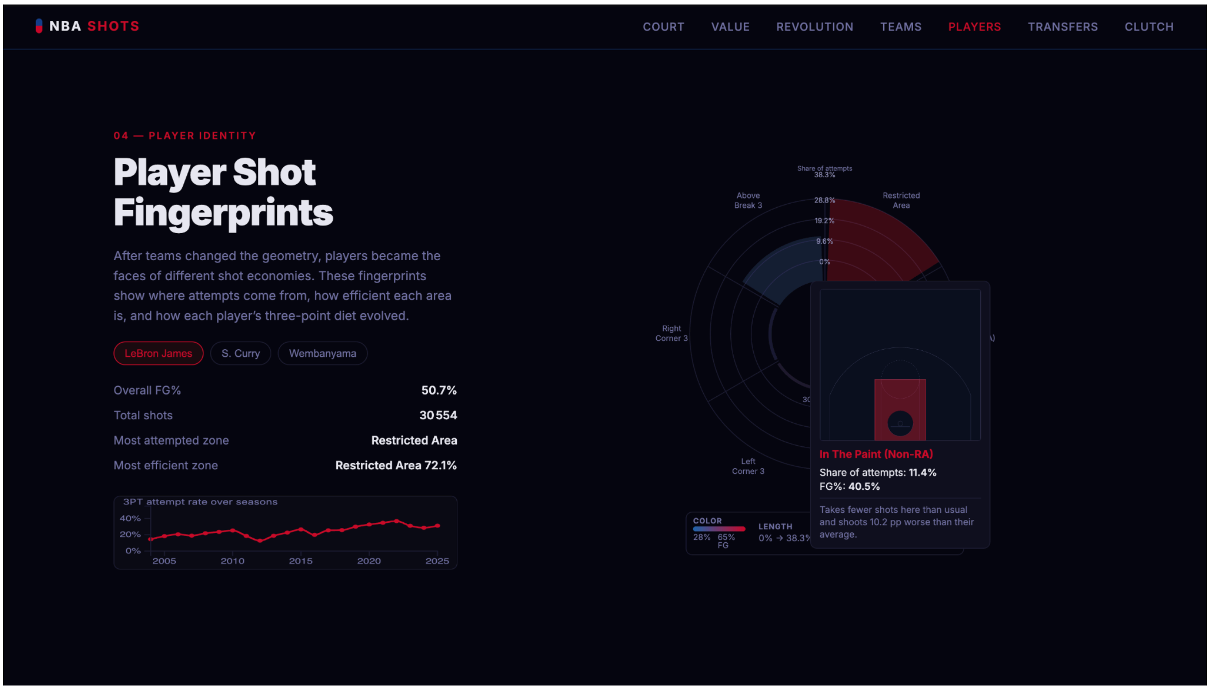
Team DNA - How Teams Followed the Math

Pick a season, pick a team and click "Show Champion." A grouped bar chart compares the chosen team's shot distribution to the league average and the champion of that season. This answers which strategies actually won.
We decided to show percent-point differences next to raw percentages. This way users can read the bars or read the numbers.



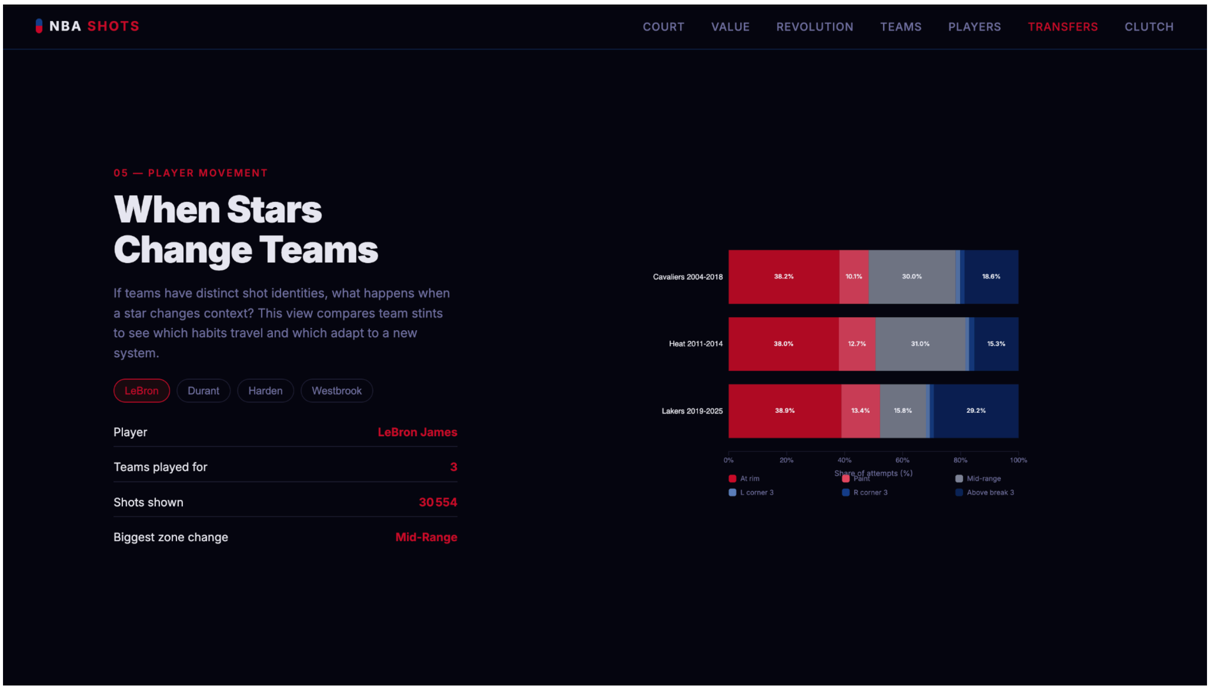
Fingerprint - Player Shot Fingerprints

A radial bar chart laid over a half-court silhouette where each wedge corresponds to a court zone. We decided to show three players: LeBron James, Stephen Curry, Victor Wembanyama.
We used a radial chart as the half-court is roughly semicircular. A polar layout matches that geometry, and the resulting shape is the player's signature. This allows for instant comparisons across players. Below it sits a mini line chart tracking three-point share per season.



Player Evolution - When Stars Change Teams

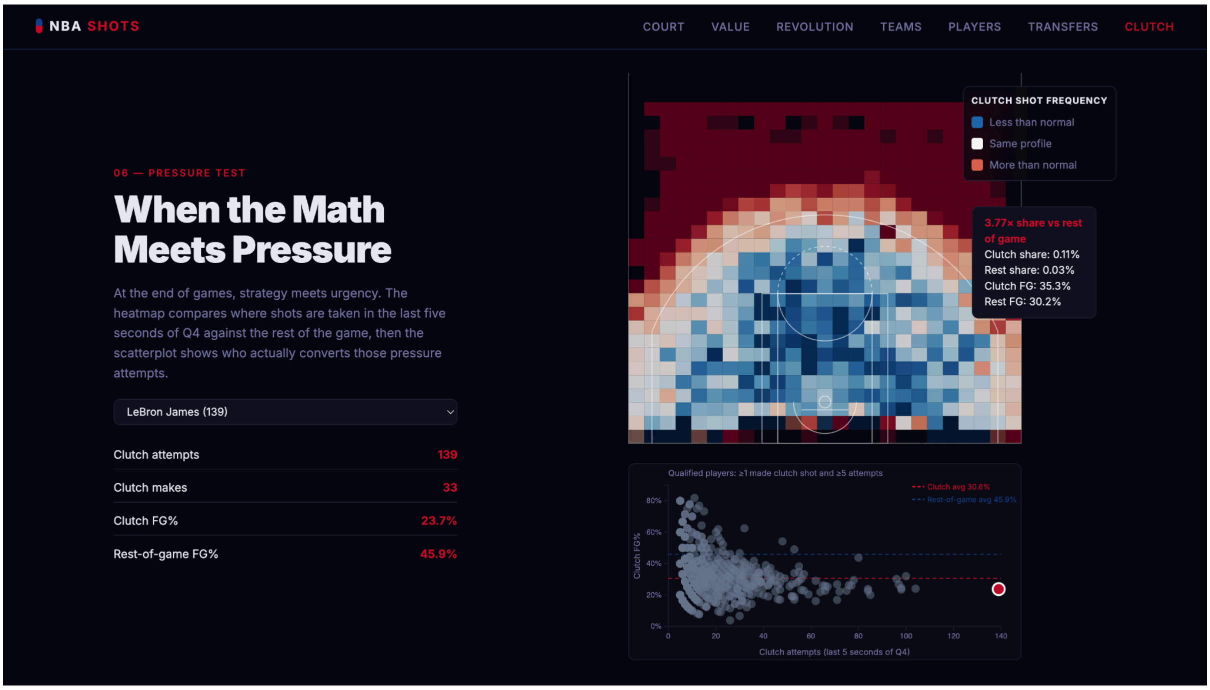
Multi-stint comparison for four high-profile movers: LeBron, Durant, Harden, Westbrook. Each stint is one row and stacked bars show how the player's shots shifted between franchises. This tests whether shot identity is the player or the system.



Clutch - When the Math Meets Pressure

Two linked views: a hexbin map of shot frequency in the last 5 seconds of Q4 relative to the rest of the game to answer whether behaviour changes under pressure, and a scatter plot of every qualified clutch shooter to plot attempts against FG%.

We decided to use the same divergent palette for the clutch heatmap than for the Revolution part, but with a different baseline (rest-of-game shot profile).



Design language & visual choices

Colour

- NBA red (#C9082A) and NBA blue (#17408B) as accent colours throughout. They are the league's colours, instantly readable to the target audience.
- Divergent red-blue (RdBu) is used for comparative heatmaps to show frequency vs league average, clutch vs rest. Red = more, blue = less.
- Sequential reds for FG% on the fingerprint. As efficiency is one-directional (higher is better) a single-hue ramp is a good choice.
- Dark background throughout gives the colour-coded marks more contrast and gives the site a warmer feel than a Wikipedia page.

Mark and channel choices

We mapped each variable to the channel that survives perceptual decoding best:

- Position for shot location.
- Hue for direction of deviation.
- Saturation/value for magnitude of deviation.
- Length of radial bars for shot frequency.

Technical implementation

Stack

- D3.js v7: every static and interactive visualisation except the hero.
- Three.js v0.160: hero particle scene with custom GLSL shaders.
- Vanilla JS, no framework: the site is a single HTML file plus eight small JS modules. No build step, no bundler. Loads in any modern browser by serving the website/ folder.
- Python (pandas + numpy) for offline preprocessing: turns ~5 GB of raw CSVs into ten compact JSON files (~6 MB total).

Data pipeline

The 4.4 M-shot dataset cannot be served directly to a browser. Our preprocess.py script aggregates the raw seasonal CSVs into the pre-computed structures the site actually needs:

- hex_by_season.json: hexbin counts + FG% per (season, hex cell)
- expected_value.json: grid-cell EV / FG% / volume
- players_fingerprint.json, player_team_evolution.json
- team_profiles.json, champion_profiles.json
- clutch_shots.json + clutch_comparison.json
- hero_particles.json, trajectory_samples.json

This shifts the heavy lifting off the client. The browser loads only what the current viewport needs and can stay smooth even on a laptop.

Code organisation

website/js/court.js is the shared utility module — court drawing, coordinate transforms, formatters, team-name shortening. Each section has its own self-contained file (value.js, revolution.js, …) that wraps its visualisation in an async IIFE. This kept us out of dependency-injection complexity while still avoiding global namespace pollution.

Challenges

Scale. The biggest single technical problem was that 4.4 M shots will never fit through a fetch into a browser. We solved it by treating the visualisation layer and the data layer as two separate problems: heavy aggregation in Python, lightweight rendering in JS.
Court coordinate system. The raw data uses feet (basket at origin) but SVG uses pixel coordinates (origin top-left). Every visualisation needed the same projection. We centralised it in Court.toX / Court.toY so a single source of truth controlled all seven visualisations.
Hero performance. Animating 75,000 particles with per-particle delays and per-year masking required moving the animation off the CPU. We wrote a custom GLSL shader (hero.js:4-50) that does all interpolation on the GPU; the JS side only updates a single uProgress uniform per frame.
Telling one story out of seven visualisations. Our first draft had seven good charts and no narrative — a dashboard. The fix was editorial, not technical: numbered section tags, explicit handoff sentences between sections, and a deliberate ordering (premise → response → zoom → test). This is where Lectures 7.1 (Designing for Data Viz) and 12.1 (Storytelling) paid off most.
Franchise relocations and renames. Twenty-two seasons cover Seattle → OKC, New Jersey → Brooklyn, Charlotte Bobcats → Hornets, and several others. We standardised to current franchise identity for team-level views, while preserving historical names in the raw clutch / fingerprint records so the labels match the era.
Colour on dark background. Standard RdBu diverging palettes flicker against pure black. We softened the background to #0d0e12 and brightened the saturated end of the palette — small change, big readability win.

Peer assessment

This project was a collaborative effort. The concept, narrative arc, and major design decisions were agreed on together, and the Python preprocessing pipeline, essential since 4.4 million shots can't be served to a browser directly, was a shared responsibility, with all three of us designing the pre-computed structures and verifying the numbers behind each visualisation.
The visualisations were divided as follows. Lysandre Costes led the exploratory data analysis and built the Player Evolution and Clutch sections. Matisse Van Schalkwijk built the Hero particle scene, the Value heatmap, and the Revolution animated hexbin section. Cléo Renaud built the Team DNA and Fingerprint sections and led the process book.
The shared court utilities, colour language, narrative structuring, and debugging were done jointly. We consider the three of us to have contributed equally.